Chapter 1. The Genesis of Fourier Analysis

The vibrating string and the flow of heat

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1.1 The vibrating string

Motivation. The vibrating string and heat flow lead to linear PDEs whose separated solutions are sines and cosines, so the general solution is a Fourier series; which functions admit one?

Simple harmonic motion. A mass m on a spring with constant k obeys $-ky = my''$. With $c = \sqrt{k/m}$, $y(t) = a\cos ct + b\sin ct = A\cos(ct - \varphi)$: **amplitude** $A = \sqrt{a^2 + b^2}$, **period** $2\pi/c$, **frequency** $c/2\pi$, $a = y(0)$, $b = y'(0)/c$. Euler: $e^{ix} = \cos x + i\sin x$, $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

Standing and travelling waves. A **travelling wave** is $u(x,t) = F(x - ct)$ (a profile moving at speed c); a **standing wave** is $u(x,t) = \varphi(x)\psi(t)$ (fixed shape, varying amplitude).

1.1.1 Derivation of the wave equation

Model a string of length L, density ρ, tension τ, fixed at its ends, by N particles of mass ρh ($h = L/N$) at $x_n = nh$ joined by springs, with vertical displacements $y_n(t)$. Hooke's law for the vertical tension components gives

$$\rho h\, y_n''(t) = \frac{\tau}{h}\big(y_{n+1}(t) + y_{n-1}(t) - 2y_n(t)\big).$$

As $h \to 0$, $y_n(t) \to u(x,t)$, the right side over h tends to $\tau\,\partial_x^2 u$: the **wave equation**

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\tau/\rho}. \tag{1.1}$$

(Rescaling $x \mapsto x/L$, $t \mapsto ct/L$ gives $c = 1$, $L = \pi$.) It is linear and invariant under translations in x and t.

1.1.2 Solution to the wave equation

d'Alembert's solution. With $\xi = x + t$, $\eta = x - t$, $v(\xi,\eta) = u(x,t)$, the equation becomes $\frac{\partial^2 v}{\partial\xi\partial\eta} = 0$, so $v = F(\xi) + G(\eta)$:

$$u(x,t) = F(x+t) + G(x-t),$$

two travelling waves. On $\mathbb{R}$ with $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$:

$$u(x,t) = \frac{f(x+t) + f(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} g(y)\,dy. \tag{1.2}$$

Check [Supplement]: $F + G = f$, $F' - G' = g$ give $F = \frac{f}{2} + \frac{1}{2}\int_0^x g + C$, $G = \frac{f}{2} - \frac{1}{2}\int_0^x g - C$. On $[0,\pi]$ with $u(0,t) = u(\pi,t) = 0$, extend f, g oddly to $[-\pi,\pi]$ and 2π-periodically; then (1.2) satisfies the boundary conditions.

Superposition of standing waves. $u = \varphi(x)\psi(t)$ gives $\frac{\psi''}{\psi} = \frac{\varphi''}{\varphi} = \lambda$. Boundedness forces $\lambda = -m^2 \le 0$, so $\psi = A\cos mt + B\sin mt$, $\varphi = \tilde{A}\cos mx + \tilde{B}\sin mx$, and $\varphi(0) = \varphi(\pi) = 0$ forces $\varphi = \sin mx$, $m = 1, 2, \ldots$:

$$u_m(x,t) = (A_m\cos mt + B_m\sin mt)\sin mx,$$

the m-th **harmonic** ($m = 1$ **fundamental tone**, $m \ge 2$ **overtones** of frequency $m/2\pi$). By linearity

$$u(x,t) = \sum_{m=1}^{\infty}(A_m\cos mt + B_m\sin mt)\sin mx \tag{1.3}$$

is a candidate, and $u(x,0)=f(x)$ requires

$$f(x)=\sum_{m=1}^{\infty} A_m \sin mx. \tag{1.4}$$

Fourier's question: can every reasonable f on $[0,\pi]$ with $f(0)=f(\pi)=0$ be written as (1.4)? If so, $\int_0^\pi \sin mx \sin nx\,dx = \frac{\pi}{2}\delta_{mn}$ forces

$$A_m=\frac{2}{\pi}\int_0^\pi f(x)\sin mx\,dx,$$

and B_m is fixed by $\partial_t u(x,0)=g(x)=\sum mB_m \sin mx$. Extending f oddly, (1.4) is a special case of

$$F(x)=\sum_{m=-\infty}^{\infty} a_m e^{imx}, \qquad a_m=\frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)e^{-imx}\,dx. \tag{1.5}$$

1.1.3 Example: the plucked string

Pluck at $p\in(0,\pi)$ to height h, zero initial velocity: $f(x)=\frac{xh}{p}$ on $[0,p]$, $f(x)=\frac{h(\pi-x)}{\pi-p}$ on $[p,\pi]$, $g=0$. Then $B_m=0$ and, by parts,

$$A_m=\frac{2h\sin mp}{m^2 p(\pi-p)}, \qquad u(x,t)=\sum_{m\geq 1} A_m \cos mt \sin mx.$$

As $\sum |A_m|<\infty$, the series converges uniformly and u is continuous. In general (1.3) need not be twice differentiable, so it solves the wave equation only in a generalised sense (compare d'Alembert).

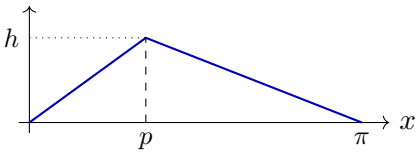


Figure. *The plucked string at $t=0$.*

1.2 The heat equation

Motivation. Steady heat flow in a disc, separated in polar coordinates, leads to Fourier series in the angle θ.

1.2.1 Derivation of the heat equation

For a plate $\mathbb{R}^2$ with temperature $u(x,y,t)$, specific heat σ and conductivity κ, the heat in a small square S is $\sigma\iint_S u$ and the flux across ∂S is $\kappa\int_{\partial S}\partial u/\partial n$ (Fourier's law). Equating the rate of change with the flux, dividing by the area and shrinking S:

$$\frac{\partial u}{\partial t}=\frac{\kappa}{\sigma}\Big(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\Big)=\frac{\kappa}{\sigma}\,\triangle u. \tag{1.6}$$

A **steady state** ($\partial u/\partial t=0$) satisfies the **Laplace equation** $\triangle u=0$; its solutions are **harmonic functions**.

1.2.2 Steady-state heat equation in the disc

Dirichlet problem. Find u with $\triangle u = 0$ in $D = \{x^2 + y^2 < 1\}$ and $u = f$ on the circle C. In polar coordinates

$$\triangle u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$$

$u = F(r)G(\theta)$ gives $\frac{r^2F'' + rF'}{F} = -\frac{G''}{G} = \lambda$. G is 2π-periodic, so $\lambda = m^2$, $G = e^{\pm im\theta}$; $r^2F'' + rF' - m^2F = 0$ gives $F = r^{\pm m}$ $(m \geq 1)$ or $1, \log r$ $(m = 0)$, and boundedness at 0 leaves $F = r^{|m|}$:

$$u(r,\theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}, \qquad f(\theta) = \sum a_m e^{im\theta}. \tag{1.7}$$

Inserting $a_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\varphi)e^{-im\varphi}d\varphi$ and summing formally,

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi) f(\varphi)\, d\varphi, \qquad P_r(\gamma) = \sum_{m\in\mathbb{Z}} r^{|m|}e^{im\gamma} = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2},$$

the **Poisson kernel** (made rigorous in Chapter 2).

1.3 Selected exercises

Motivation. Book exercises justifying the separated solutions, the discrete-to-continuous limit, the polar Laplacian and the plucked-string coefficients.

Exercise 1.6 (Solutions of $f'' + c^2 f = 0$)

Let $c > 0$ and let f be a C^2 function on $\mathbb{R}$ with $f''(t) + c^2 f(t) = 0$. Prove that there are constants a, b with $f(t) = a\cos ct + b\sin ct$ for all t.

Solution.

Step 1. Let $g(t) = f(t)\cos ct - c^{-1}f'(t)\sin ct$ and $h(t) = f(t)\sin ct + c^{-1}f'(t)\cos ct$, both C^1. Using $f'' = -c^2 f$,

$$g'(t) = f'\cos ct - cf\sin ct - c^{-1}f''\sin ct - f'\cos ct = -cf\sin ct + cf\sin ct = 0,$$
$$h'(t) = f'\sin ct + cf\cos ct + c^{-1}f''\cos ct - f'\sin ct = cf\cos ct - cf\cos ct = 0.$$

Step 2. So $g \equiv a = f(0)$ and $h \equiv b = f'(0)/c$ are constant.

Step 3. Then, for every t,

$$a\cos ct + b\sin ct = f(\cos^2 ct + \sin^2 ct) + c^{-1}f'(-\sin ct\cos ct + \sin ct\cos ct) = f(t).$$

This justifies ψ and φ in §1.1.2. ∎

Exercise 1.8 (Second difference quotient)

Let F be C^2 on (a,b) and $x \in (a,b)$. Show that for $x + h \in (a,b)$

$$F(x+h) = F(x) + hF'(x) + \tfrac{h^2}{2}F''(x) + h^2\varphi(h), \qquad \varphi(h) \to 0 \ (h \to 0),$$

and deduce that $\big(F(x+h) + F(x-h) - 2F(x)\big)/h^2 \to F''(x)$.

Solution.

Step 1. $F(x+h)-F(x)=\int_x^{x+h}F'(y)\,dy$, $F'(y)=F'(x)+\int_x^y F''(s)\,ds$; write $F''(s)=F''(x)+(F''(s)-F''(x))$ and use $\int_x^{x+h}(y-x)\,dy=h^2/2$:

$$F(x+h)=F(x)+hF'(x)+\tfrac{h^2}{2}F''(x)+R(h),\qquad R(h)=\int_x^{x+h}\!\!\int_x^{y}\big(F''(s)-F''(x)\big)\,ds\,dy.$$

Step 2. Let $\omega(\delta)=\sup_{|s-x|\le\delta}|F''(s)-F''(x)|\to 0$ (F'' continuous). The inner integral is at most $|y-x|\,\omega(|h|)$, so $|R(h)|\le\frac{h^2}{2}\omega(|h|)$ for either sign of h.

Step 3. With $\varphi(h)=R(h)/h^2$, $|\varphi(h)|\le\frac12\omega(|h|)\to 0$.

Step 4. Adding the expansions at $\pm h$, the first-order terms cancel:

$$\frac{F(x+h)+F(x-h)-2F(x)}{h^2}=F''(x)+\varphi(h)+\varphi(-h)\to F''(x).$$

This is the limit giving (1.1). ∎

Exercise 1.9 (Missing harmonics of the plucked string)

For the plucked string of §1.1.3, derive $A_m=\frac{2h\sin mp}{m^2p(\pi-p)}$. For which p are the 2nd, 4th, 6th, … harmonics missing? For which p are the 3rd, 6th, 9th, … harmonics missing?

Solution.

Step 1. f is continuous, piecewise linear, $f(0)=f(\pi)=0$, so by parts

$$\int_0^\pi f(x)\sin mx\,dx=\frac1m\int_0^\pi f'(x)\cos mx\,dx=\frac1m\Big(\frac{h}{p}\int_0^p\cos mx\,dx-\frac{h}{\pi-p}\int_p^\pi\cos mx\,dx\Big).$$

Step 2. The integrals are $\pm\frac{\sin mp}{m}$, so the bracket is $\frac{h\sin mp}{m}\big(\frac1p+\frac{1}{\pi-p}\big)=\frac{\pi h\sin mp}{m\,p(\pi-p)}$; multiply by $\frac{2}{\pi}$.

Step 3. $A_m=0$ iff $\sin mp=0$. All even harmonics vanish iff $\sin 2kp=0$ for all k; $k=1$ forces $p=\pi/2$, which conversely works. (Odd ones remain: $\sin(m\pi/2)=\pm1$.)

Step 4. Multiples of 3 vanish iff $\sin 3p=0$, i.e. $p=\pi/3$ or $2\pi/3$. ∎

Exercise 1.10 (Laplacian in polar coordinates)

Let u be C^2 on an open set of $\mathbb{R}^2\setminus\{0\}$ and $v(r,\theta)=u(r\cos\theta,r\sin\theta)$. Prove

$$u_{xx}+u_{yy}=v_{rr}+\frac1r v_r+\frac{1}{r^2}v_{\theta\theta},\qquad |u_x|^2+|u_y|^2=|v_r|^2+\frac{1}{r^2}|v_\theta|^2.$$

Solution. Write $c=\cos\theta$, $s=\sin\theta$; derivatives of u are taken at (rc,rs).

Step 1 (first derivatives). By the chain rule $v_r=cu_x+su_y$ and $v_\theta=-rsu_x+rcu_y$.

Step 2 (gradient identity). Hence $\frac1r v_\theta=-su_x+cu_y$ and

$$|v_r|^2+\tfrac{1}{r^2}|v_\theta|^2=|cu_x+su_y|^2+|-su_x+cu_y|^2=(c^2+s^2)(|u_x|^2+|u_y|^2),$$

the cross terms $\pm2cs\,\mathrm{Re}(u_x\bar u_y)$ cancelling.

Step 3 (second derivatives). Differentiating Step 1, with $u_{xy}=u_{yx}$,

$$v_{rr}=c^2u_{xx}+2csu_{xy}+s^2u_{yy},$$
$$v_{\theta\theta}=-rcu_x-rsu_y+r^2(s^2u_{xx}-2csu_{xy}+c^2u_{yy})=-rv_r+r^2(s^2u_{xx}-2csu_{xy}+c^2u_{yy}).$$

Step 4. So $v_{rr}+\frac1r v_r+\frac{1}{r^2}v_{\theta\theta}=(c^2+s^2)(u_{xx}+u_{yy})$. ∎

Exercise 1.11 (The Euler equation $r^2F'' + rF' - n^2F = 0$)

Let $n \in \mathbb{Z}$. Show that the twice differentiable solutions on $r > 0$ of $r^2F''(r)+rF'(r)-n^2F(r) = 0$ are exactly the linear combinations of r^n, r^{-n} if $n \neq 0$, and of $1, \log r$ if $n = 0$.

Solution. Only n^2 enters, so let $n \geq 0$.

Step 1 (these are solutions). $F = r^{\pm n}$: $r^2(\pm n)(\pm n - 1)r^{\pm n-2} + r(\pm n)r^{\pm n-1} - n^2r^{\pm n} = (n^2 \mp n \pm n - n^2)r^{\pm n} = 0$. $F = \log r$: $r^2(-r^{-2}) + r \cdot r^{-1} = 0$; constants trivially. Combinations by linearity.

Step 2 (reduction). For a solution F let $g = r^{-n}F$. From $F = r^n g$,

$$r^2F'' + rF' - n^2F = r^n\big[r^2g'' + 2nrg' + n(n-1)g + rg' + ng - n^2g\big] = r^{n+1}\big[rg'' + (2n+1)g'\big],$$

so $rg'' + (2n+1)g' = 0$ on $(0, \infty)$.

Step 3. Then $\big(r^{2n+1}g'\big)' = r^{2n}\big(rg'' + (2n+1)g'\big) = 0$, so $r^{2n+1}g' = C$.

Step 4. $n \geq 1$: $g = -\frac{C}{2n}r^{-2n} + D$, $F = Dr^n - \frac{C}{2n}r^{-n}$. $n = 0$: $F = g = C\log r + D$. This is the classification used in §1.2.2 (and in Theorem 5.7). ∎